

Homework 5

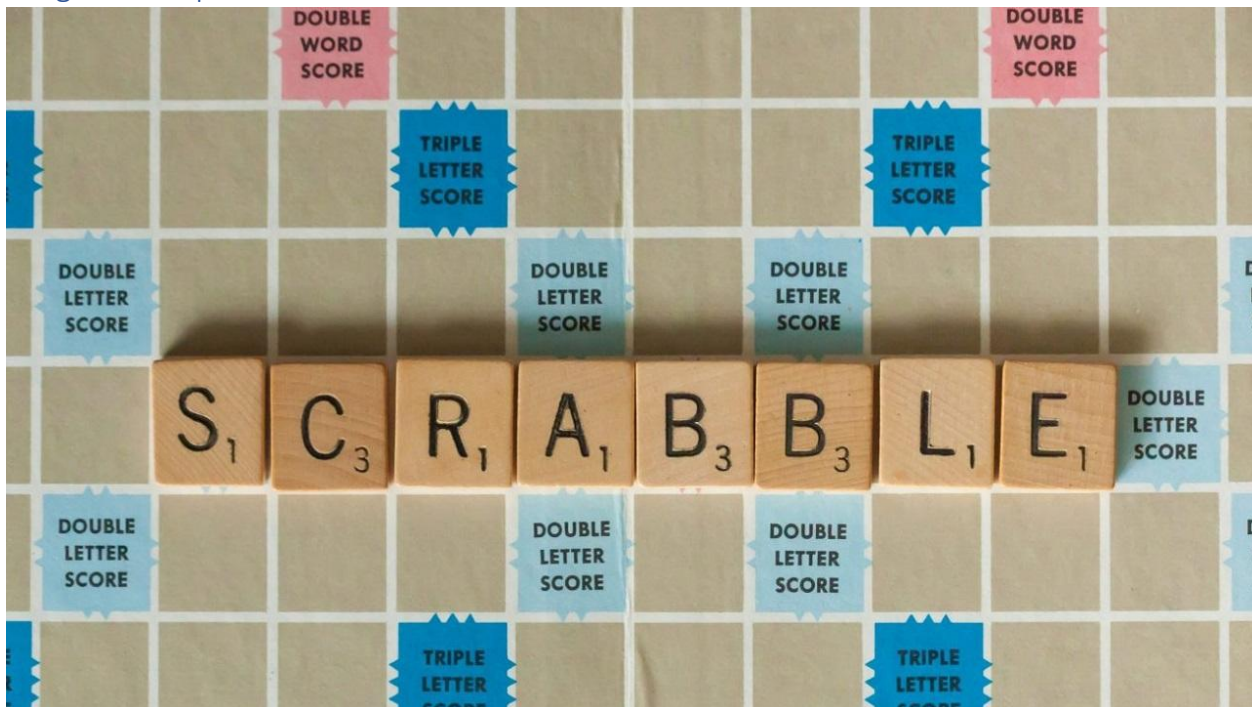
“React and React Router”

The following problem set is worth 100 points total. Please submit a zip file containing all relevant code to the Canvas drop box by the due date of the assignment.

Requirements

This assignment involves creating a React application using **React Router** and games that can be implemented with **drag-and-drop components**.

Drag-and-Drop Game #1 – Word Scrambler



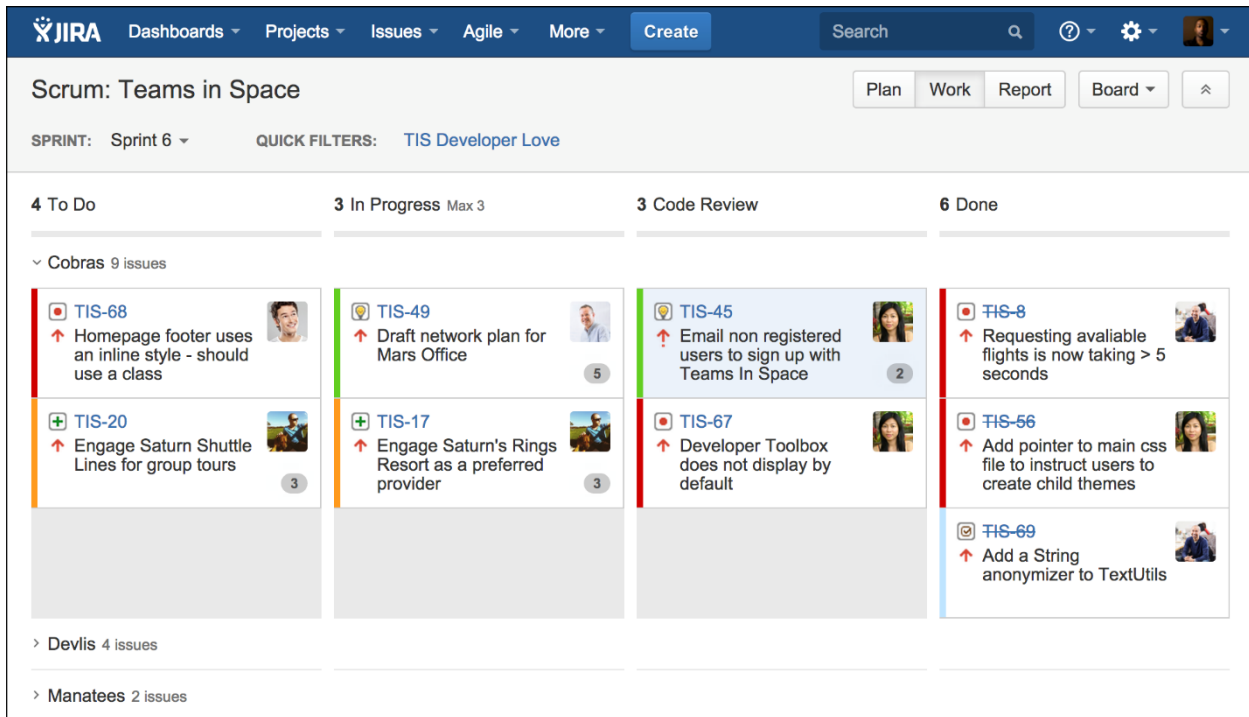
The game *Scrabble* is a classic game where players are given tiles corresponding to letters in the alphabet and the objective of the game is to use the letters to form words in the dictionary. Obviously the more complicated the word formed the higher the score.

This first game is somewhat inspired by the classic game and can be simplified as follows:

1. Select a word randomly from a dictionary (a list of words you have stored in the code base)
2. Randomize the letters in the word
3. Display the letters as tiles that the user can drag-and-drop into place
4. The game is won when the user can formulate the hidden word
5. Ask the user if another game is to be played

Drag-and-Drop Game #2 – JIRA Clone

A well-known piece of software that product teams use today is JIRA, developed by Atlassian. JIRA is a tool that leads product owners organize tasks for developers to be completed in a short period of time (call it a **sprint**). One of the most prominent features in JIRA is the ability to organize tasks into columns known as “**swim lanes**”:



With React, create a simple JIRA clone that mirrors the screen shot above:

- Produce four “swim lanes”: To Do, In Progress, Code Review and Done
- Add a button on the form, allowing the user to create a new task
 - In the case of JIRA, a ticket number is automatically created, e.g. TIS-76, where TIS is the project key and 76 is an auto-incrementing number
 - A red upward arrow or a green downward arrow designates task importance
- Allow the user to drag the initially created task, which starts at To Do through the four “swim lanes”
- When the task hits the Done column, strike-through the ticket number on the board

Drag-And-Drop Game #3 – Color Blender

Create a React component designed as follows:



Your component will have two zones, a drag zone with six colors: red, blue, yellow, green, orange and purple. Each of these colors can be dragged to one of two drop zones. When both drop zones are populated with colors, show the user the mixing result (as an example red and blue mixed together forms purple). Provide functionality for the user to be able to reset the choices, or possibly clear out a drop zone by removing a color out of it.

React Router

Each of these three games should be rendered using **React Router**, as tabs on a navigation bar. For handling drag-and-drop functionality, several solutions exist. The most interesting and elegant solution is Pragmatic Drag and Drop, a library created oddly enough created by Atlassian (the creators behind JIRA): <https://atlassian.design/components/pragmatic-drag-and-drop/about>

Submission Requirements

For submission, upload a zip file containing all assets I need to run your application. Do not include the **node_modules** directory in your submission.

Submit using the Canvas drop box before the due date.